

Chapter 4- BIOS, Booting, Partitioning & Recovery

◆ 1. What is BIOS? (Basic Input Output System)

👉 **BIOS** is a firmware (software on motherboard) that controls hardware during startup.

- You can see system info using **msinfo32** in Windows.
- BIOS is stored on the **motherboard chip**, not inside Windows.

⚙️ Main Functions of BIOS

- Controls **input/output devices** (keyboard, mouse, disk)
- Starts the **boot process**
- Checks hardware health

🔍 Daily-life Example

Like a **school principal** who checks everything before classes start.

◆ 2. Types of BIOS

● 1. Legacy BIOS

- Introduced around **2000**
- Supports max **2TB hard disk**
- Works with **MBR partition**
- No mouse support (keyboard only)

● 2. UEFI (Unified Extensible Firmware Interface)

- Modern BIOS (replaced legacy)
- Supports **more than 2TB disks**
- Supports **GPT partition**

- Has **mouse + keyboard support**
- Faster and more secure

Daily-life Example

- Legacy = Old keypad phone 📞
- UEFI = Modern smartphone 📱

◆ **3. BIOS Boot Process (Important Flow)**

When system starts:

1. BIOS runs **POST (Power-On Self Test)** 🧪
2. Finds **boot device** (HDD/SSD/USB)
3. Loads bootloader:
 - Windows → **bootmgr**
 - Linux → **GRUB loader**
4. Bootloader loads OS into RAM
5. Login screen appears

⚠️ If boot files are deleted →

👉 ❌ **Error: "No Bootable Device Found"**

◆ **4. Hard Disk Partitioning**

★ **What is Partitioning?**

Dividing a physical disk into multiple parts:

- C: → OS
- D: → Personal files
- E: → Games/work

⚙️ Open Disk Management

👉 Windows + R → diskmgmt.msc

💡 IT Rule

Counting starts from **0**

👉 First disk = **Disk 0**

🔍 Daily-life Example

Like dividing a **cupboard into shelves**

◆ 5. Partition Types: MBR vs GPT

● MBR (Master Boot Record)

- Introduced in **1983**
- Max disk support: **2TB**
- Allows:
 - 3 Primary partitions
 - 1 Extended partition
 - Inside extended → Logical partitions

📌 Storage Location

Stored in **first sector of disk**

⚠️ Disadvantage

- If first sector is corrupted → data loss
- Recovery success ~70%

Boot Identification

 Straight loading line = MBR

Example

Like storing all data in **one notebook**

GPT (GUID Partition Table)

- Modern and advanced system

Advantages:

- Supports **>2TB disks**
- Up to **128 partitions**
- No need for extended/logical partitions
- Multiple backup copies
- Safer & faster recovery

Boot Identification

 Spinning circle = GPT

Example

Like saving files with **multiple backups**

6. Partition Types (Important for IT Support)

1. Primary Partition

- Main partition for OS & data
- Required for OS installation

 Example: **C: Drive**

◆ 2. Extended Partition (MBR only)

- Works as a container
 - Only **1 allowed**
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◆ 3. Logical Partition

- Inside extended partition
- Cannot boot OS

👉 Example: D:, E:, F:

◆ 4. Active Partition

- Contains boot info
- Only **1 active partition allowed**

👉 Example: System Reserved

◆ 5. System Partition

- Contains boot files:
 - bootmgr
 - BCD
 - EFI folder

👉 Loads OS into RAM

◆ 7. EFI System Partition (ESP)

👉 Used only in **UEFI systems**

📁 **Contains:**

- bootmgr.efi
- EFI boot files

- Drivers

Purpose:

- Controls boot process
- Loads login screen

 Without ESP → System won't boot

8. Recovery Partition

 Hidden partition created by Windows

Uses:

- System repair
- Reset PC
- Troubleshooting
- Backup restore

Contains:

- Windows Recovery Environment (WinRE)

Example

Like a first-aid kit  for your PC

9. Entering Recovery Mode

Method 1 (Settings)

Settings → System → Recovery → Advanced Startup

Method 2 (Force Shutdown)

- Restart system
- Force shutdown 5–6 times
- Auto recovery starts

◆ 10. Boot Flow (Full Understanding)

1. BIOS/UEFI starts
 2. System partition loads boot files
 3. Boot manager loads OS
 4. Login screen appears
 5. If OS fails → Recovery starts
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◆ 11. BIOS vs UEFI Check

■ Method 1: Boot Screen

- Straight line → Legacy
- Spinning circle → UEFI

■ Method 2: BIOS Interface

- Mouse works → UEFI
 - Only keyboard → Legacy
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◆ 12. BIOS & Boot Keys (Important for Support)

Brand BIOS Key Boot Menu

Acer F2 / DEL F12 / ESC

Dell F2 F12

HP F10 ESC

Lenovo F1 / F2 F12

ASUS F2 / DEL F8

👉 Used during troubleshooting (OS install, boot issues)

◆ 13. BIOS Security Settings (HP Example)

🔒 BIOS Administrator Password

- Protects BIOS access
- Prevents unauthorized changes

◆ 14. Secure Boot (VERY IMPORTANT) 🔥

👉 Available only in UEFI

⚙️ What it does:

- Protects system from malware during boot
- Checks files before OS loads

💡 Why needed?

Problem:

- Malware loads into RAM **before antivirus starts**

Solution:

- Secure Boot scans during BIOS stage

🔍 Flow:

BIOS → Bootmgr → OS → Antivirus

👉 Secure Boot protects **before OS loads**

🧪 Example

Like checking bags **before entering airport** ✈️

◆ 15. BIOS Advanced Boot Options

🔌 USB Boot Issue (Common Problem)

👉 Problem:
Pendrive not detecting

👉 Solution:

- Go to BIOS → Boot Options
 - Enable **USB Boot**
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1234 Boot Order Priority

Example:

1. SSD
 2. USB
 3. Network Boot
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💻 Types of SSD

- SATA SSD
 - NVMe SSD
 - M.2 SSD
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🌐 Network Boot (Advanced Concept)

- Used in companies/malls
 - OS loads via **Ethernet from server**
 - No hard disk needed
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◆ 16. Windows 11 ISO Download

1. Go to Microsoft website
 2. Select:
 - Windows 11 ISO (x64)
 - Multi-edition
 3. Choose language
 4. Download 64-bit ISO
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◆ 17. Rufus Tool

👉 Used to create **bootable USB**

- Download from official site
 - File: rufus-x.x.exe
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🎯 IT Support Practical Tips (VERY IMPORTANT)

✓ Always check:

- Boot order
- Partition type (MBR/GPT)
- BIOS mode (UEFI/Legacy)
- Secure Boot status

✓ Common errors:

- No boot device → boot files missing
- USB not detecting → USB boot disabled
- OS not installing → wrong partition type

✓ Fast troubleshooting mindset:

- Identify problem quickly

- Check BIOS first
 - Then disk & bootloader
 - Then OS
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Final Understanding

- 👉 BIOS = Starts system
 - 👉 Bootloader = Loads OS
 - 👉 Partition = Organizes disk
 - 👉 Secure Boot = Protects startup
 - 👉 Recovery = Fixes system
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Chapter 4 - SUMMARY

BIOS (Basic Input Output System)

-  BIOS is firmware stored on motherboard chip
-  It starts the system and checks hardware

Functions:

- Hardware check (POST) 
 - Start boot process 
 - Manage input/output devices 
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Legacy vs UEFI

Feature	Legacy BIOS 	UEFI 
Disk support	Up to 2TB	More than 2TB
Partition	MBR	GPT
Interface	Keyboard only	Mouse + GUI
Speed	Slow	Fast

-  UEFI = Modern & secure 
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Booting Process

Step-by-step:

1. Power ON 
2. BIOS/UEFI starts 
3. POST check 
4. Boot device detected 

5. Bootloader loads OS
6. Login screen appears

Partitioning

👉 Disk ko parts me divide karna:

- C: → OS
- D: → Data

💡 Example: Cupboard shelves

● **MBR vs GPT (VERY IMPORTANT)**

● **MBR**

- Max 2TB
- 4 partitions
- Risky (single copy)

● **GPT**

- 2TB
- 128 partitions
- Backup available

👉 GPT is safer & modern 

◆ **Partition Types**

- Primary → OS install
- Extended → Container
- Logical → Inside extended
- Active → Boot partition
- System → Boot files

◆ EFI Partition (UEFI)

👉 Contains:

- bootmgr.efi
- Boot files

👉 Without EFI → System will NOT boot ❌

🔧 Recovery Partition

👉 Hidden partition for:

- Reset PC
- Troubleshooting
- Repair

💡 Like first aid kit 🩹

🔒 Secure Boot (VERY IMPORTANT)

👉 Works in UEFI only

👉 Function:

- Blocks malware during boot
 - Verifies files before OS loads
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🔧 Boot Issues (Common)

❌ No Boot Device:

- Wrong boot order
- OS missing
- Disk failure

✖ USB Not Detect:

- USB boot disabled
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🌐 Advanced Concepts

👉 Network Boot:

- OS loads from server

👉 Rufus:

- Creates bootable USB
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🎯 CONCLUSION

👉 Full boot flow:

BIOS → Bootloader → OS → Login

👉 Core understanding:

- BIOS = Start system
- Partition = Organize disk
- Bootloader = Load OS
- Recovery = Fix issues

👉 Final line:

“If you understand booting deeply, you can fix 80% system problems” 🔥

Q & A

◆ BIOS QUESTIONS

? Q1: What is BIOS?

👉 Answer:

BIOS is firmware stored on the motherboard that initializes hardware and starts the booting process.

It acts as a bridge between hardware and operating system.

? Q2: What is POST?

👉 Answer:

POST (Power-On Self Test) is a process where BIOS checks hardware components like RAM, CPU, and storage before booting.

? Q3: Where is BIOS stored?

👉 Answer:

BIOS is stored in a chip on the motherboard, not in the operating system.

◆ UEFI vs LEGACY QUESTIONS

? Q4: Difference between BIOS and UEFI?

👉 Answer:

UEFI is an advanced version of BIOS with better speed, security, and support for large disks (>2TB).

? Q5: How to identify UEFI or Legacy?

👉 Answer:

- Mouse works → UEFI
 - Only keyboard → Legacy
 - Spinning circle boot → UEFI
 - Straight line → Legacy
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◆ BOOTING QUESTIONS

? Q6: What is booting?

👉 Answer:

Bootting is the process of starting a computer and loading the operating system into RAM.

? Q7: What is bootloader?

👉 Answer:

A program that loads OS:

- Windows → bootmgr
 - Linux → GRUB
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? Q8: What is boot device?

👉 Answer:

Device from which OS loads, like HDD, SSD, or USB.

◆ PARTITION QUESTIONS

? Q9: What is partitioning?

👉 Answer:

Dividing a hard disk into multiple sections to organize OS and data.

? Q10: What is MBR?

👉 Answer:

Master Boot Record is an old partitioning system that supports up to 2TB disk and 4 partitions.

? Q11: What is GPT?

👉 Answer:

GUID Partition Table is a modern partitioning system that supports large disks and multiple partitions with backup.

◆ PARTITION TYPES QUESTIONS

? Q12: What is primary partition?

👉 Answer:

Main partition where OS is installed.

? Q13: What is extended partition?

👉 Answer:

A container that holds logical partitions (only in MBR).

? Q14: What is logical partition?

👉 Answer:

Partitions inside extended partition used for storing data.

? Q15: What is active partition?

👉 Answer:

Partition that contains boot information and is used to start OS.

◆ EFI & RECOVERY QUESTIONS

? Q16: What is EFI partition?

👉 Answer:

EFI System Partition contains boot files used by UEFI to start the OS.

? Q17: What is recovery partition?

👉 Answer:

A hidden partition used to repair, reset, or troubleshoot the system.

◆ SECURE BOOT QUESTIONS

? Q18: What is Secure Boot?

👉 Answer:

A security feature in UEFI that prevents unauthorized or malicious software from loading during boot.

◆ TROUBLESHOOTING QUESTIONS (VERY IMPORTANT)

BOOT ISSUE

? Q19: No Bootable Device Found?

👉 Answer:

Possible causes:

- Wrong boot order
- OS missing
- Disk failure
- Bootloader corrupt

👉 Solution:

- Check BIOS boot order
- Repair bootloader
- Check disk

USB NOT DETECT

? Q20: USB not booting?

👉 Answer:

- Enable USB boot in BIOS
 - Check bootable USB (Rufus)
 - Change boot priority
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⚠️ OS INSTALL ISSUE

? Q21: OS not installing?

👉 Answer:

- Partition type mismatch (MBR vs GPT)
 - Secure boot issue
 - Corrupt ISO
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◆ REAL JOB SCENARIOS

? Q22: System stuck on logo?

👉 Answer:

- Bootloader issue
- OS corruption

👉 Solution:

- Use recovery mode
 - Repair startup
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? Q23: BIOS settings resetting?

👉 Answer:

👉 CMOS battery dead

? Q24: Laptop not booting after update?

👉 Answer:

👉 Boot files corrupted

👉 Use recovery or reinstall OS

◆ INTERVIEW IMPRESSION QUESTIONS

? Q25: What is full boot flow?

👉 Answer:

BIOS → POST → Bootloader → OS → Login screen

? Q26: How do you troubleshoot boot issue?

👉 Answer:

- Check BIOS
 - Check boot order
 - Check disk
 - Repair OS
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